



# Quadratic equations: what we can already solve

ANSWER KEY • FOR TEACHERS

## REMEMBER

### From grade 8

Level 1 • threshold g8.inc-quad

**1.**Solve  $2x^2 = 50$ . Give both roots, separated by a semicolon.**Answer: 5; -5**

Level 1 • threshold g8.inc-quad

**2.**Solve  $x^2 - 6x = 0$ .**Answer: 0; 6**

Level 1 • threshold g8.inc-quad

**3.**Which of these numbers is a root of  $x^2 - x - 12 = 0$ ?**A** 3**B** 12**C** -4**D**  $-3 - (-3)^2 - (-3) - 12 = 9 + 3 - 12 = 0.$ 

## WORKED EXAMPLES

### Worked examples

#### YOUR TURN

Level 1 • threshold g9.qe.1

Write  $4 - x^2 = 3x$  in the form  $ax^2 + bx + c = 0$  with  $a > 0$ . Enter  $a$ ,  $b$  and  $c$ , in this order, separated by semicolons.

#### WORKED SOLUTION

 $4 - x^2 - 3x = 0$ , so  $-x^2 - 3x + 4 = 0$ . Multiplying by  $-1$  gives  $x^2 + 3x - 4 = 0$ :  $a = 1$ ,  $b = 3$ ,  $c = -4$ .**Answer:**  $a = 1$ ,  $b = 3$ ,  $c = -4$ 

#### YOUR TURN

Level 1 • threshold g9.qe.2

Solve  $(3x + 2)(x - 5) = 0$ .**Answer:**  $x_1 = -\frac{2}{3}$ ,  $x_2 = 5$

## YOUR TURN

Level 1 · threshold

g9.qe.2

Solve  $(x - 3)^2 = 25$ .

**Answer: 8; -2**

## CHECK

### Check yourself

Level 1 · threshold

g9.qe.1

1.

Which of these are quadratic equations? Choose all that apply.

☐ A  $\frac{4}{x^2} - 1 = 0$

☐ B  $3^2x + x - 4 = 0$

☒ C  $x^2 - 7x = 0$  — It is quadratic ( $a = 1$ ,  $b = -7$ ) and incomplete, because  $c = 0$ .

☒ D  $5 - 2x^2 = 0$  — It is quadratic:  $-2x^2 + 5 = 0$ , with  $a = -2$ ,  $b = 0$ ,  $c = 5$ . The order of the terms does not matter.

Level 2 · satisfactory

g9.qe.1

2.

Which of these is an **incomplete** quadratic equation?

☐ A  $x^2 + x = 1$

☐ B  $(x - 1)^2 = 0$

☒ C  $3x^2 = 5x - 3x^2 - 5x = 0$ :  $c = 0$ .

☐ D  $2x^2 + 3x - 5 = 0$

Level 1 · threshold

g9.qe.1

3.

Write  $7 - 3x^2 = 2x$  in the form  $ax^2 + bx + c = 0$  with  $a > 0$ . Enter  $a$ ,  $b$  and  $c$ , in this order.

#### WORKED SOLUTION

$7 - 3x^2 - 2x = 0$ , so  $-3x^2 - 2x + 7 = 0$ . Multiplying by  $-1$  gives  $3x^2 + 2x - 7 = 0$ :  $a = 3$ ,  $b = 2$ ,  $c = -7$ .

**Answer:  $a = 3$ ,  $b = 2$ ,  $c = -7$**

Level 1 · threshold

g9.qe.2

4.

Solve  $(x + 6)(3x - 1) = 0$ .

**Answer:  $x_1 = -6$ ,  $x_2 = \frac{1}{3}$**

Level 1 · threshold

g9.qe.2

5.

Solve  $(x + 4)^2 = 36$ .

**Answer: 2; -10**

6.

Solve  $(x - 5)^2 = -9$ .

- ☐ A  $x = -4$ 
☒ B **no real roots** — A square is never negative, so no real number works.
 ☐ C  $x = 5$ 
☐ D  $x = 8$  or  $x = 2$

7.

Ona solves  $x(x - 3) = 4$  like this: " $x = 4$  or  $x - 3 = 4$ , so  $x = 4$  or  $x = 7$ ." What is wrong?

- ☒ A **The zero-product rule needs 0 on one side. Here it is 4, so the rule does not apply, and  $x = 7$  is not a root.** — Check:  $7 \cdot (7 - 3) = 28 \neq 4$ . And her reasoning does not show that 4 is a root either:  $4 \cdot 1 = 4$  happens to work, but not because of the rule.
 ☐ B She should divide both sides by  $x$ .
 ☐ C Nothing: 4 and 7 are both roots.
 ☐ D She should have written  $x = 0$  or  $x - 3 = 0$ .

## EXAM TASK

### A hidden square

Level 3 · basic

PUPP style · 3 points

g9.qe.2

The equation  $x^2 - 2x + 1 = k$  is given, where  $k$  is a number.

- Write the left side as the square of a binomial.
- Solve the equation when  $k = 49$ .
- For which values of  $k$  does the equation have no real roots? Explain.

#### MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

- |                                                           |      |
|-----------------------------------------------------------|------|
| a) $x^2 - 2x + 1 = (x - 1)^2$ .                           | 1 pt |
| b) $x - 1 = 7$ or $x - 1 = -7$ , so $x = 8$ or $x = -6$ . | 1 pt |
| c) $k < 0$ , because $(x - 1)^2 \geq 0$ for every $x$ .   | 1 pt |

#### WORKED SOLUTION

- a)  $x^2 - 2x + 1 = x^2 - 2 \cdot x \cdot 1 + 1^2 = (x - 1)^2$ .
- b)  $(x - 1)^2 = 49$ , so  $x - 1 = 7$  or  $x - 1 = -7$ :  $x = 8$  or  $x = -6$ . Check:  $8^2 - 2 \cdot 8 + 1 = 49$  ✓ and  $(-6)^2 - 2 \cdot (-6) + 1 = 49$  ✓.
- c) The equation is  $(x - 1)^2 = k$ . A square is never negative, so for  $k < 0$  there are no real roots. (For  $k = 0$  there is one root,  $x = 1$ ; for  $k > 0$  there are two.)

**Answer:** a)  $(x - 1)^2$ ; b)  $x = 8$  or  $x = -6$ ; c)  $k < 0$ .

## TEACHER NOTES

### For the teacher

**Pupils can say:** “I can write a quadratic equation as  $ax^2 + bx + c = 0$  and read  $a, b, c$  with their signs; I can tell a complete equation from an incomplete one; I can solve a product equal to zero and a square equal to a number.”

**Where this lesson sits.** Since V-594, the definition of a quadratic equation and the incomplete equations are grade 8 content (31.2.2). This lesson collects that knowledge, names the general form, and adds two easy types: a product of linear factors equal to 0, and  $(x + p)^2 = d$ . The hook sets up lesson 1.2: (E) became easy because it could be rewritten as (D). Can that always be done?

#### Common mistakes and what helps

- Reading  $b$  and  $c$  before moving the terms to one side ( $7 - 3x^2 = 2x$  read as  $b = 2$ ). Insist on “... = 0” first.
- Signs in the zero-product rule:  $x + 6 = 0$  gives  $x = -6$ ;  $3x - 1 = 0$  gives  $x = \frac{1}{3}$ , not 3.
- Using the rule when the right side is not 0 (item c7). Ask: “Why does the rule need a 0?”
- Forgetting the negative case in  $(x + p)^2 = d$  (c5), or “taking the root” of a negative number (c6).
- Dividing by  $x$ , which loses the root  $x = 0$  (r2). Lesson 1.7 comes back to this.
- Calling an equation incomplete just because it is short ( $x^2 + x = 1$ ), or because it is easy to solve ( $(x - 1)^2 = 0$ ).

**Key terms (EN • LT):** quadratic equation • *kvadratinė lygtis*; coefficient • *koeficientas*; constant term • *laisvasis narys*; complete / incomplete equation • *pilnoji / nepilnoji lygtis*; root • *sprendinys*; zero-product rule • *sandauga lygi nuliui, kai bent vienas dauginamasis lygus nuliui*.

**Timing (45 min):**

- Remember: 5 min.
- Hook: 3 min, one try.
- Explain: 12 min.
- Examples and “Your turn”: 13 min.
- Check c1–c6: 10 min.

c7 (level 3) and the exam task are for pupils who finish early, or for homework. The exam task links this lesson to the next ones: its left side can already be recognised as a square (lesson 1.2 makes one), and the sign of  $k$  decides the number of roots (lesson 1.5 does the same with  $D$ ).

**Programme:** 32.1.1.1 “*Kvadratinės lygtys*”, bridging from 31.2.2 “*Apibrėžiamos sąvokos: antrojo laipsnio (kvadratinė) lygtis su vienu nežinomuoju ir jos sprendinys.*” NŠA threshold minimum (2025): 2.2.1 (incomplete equations) and 2.3.2 (“Which equations are quadratic?”, like c1). Achievements A1, B2.